

WEEK 4



Player Events

Code that reacts to you

플레이어의 행동에 코드가 저절로 반응해요

BIG PICTURE

Traps and rewards 📺



No chat command needed — an **event** fires by itself.

건거나 블록을 놓을 때 자동 실행

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What's an event ?

Code that runs **by itself** when the player does something.

```
def on_walk():  
    if agent.inspect(DOWN) == GOLD_BLOCK:  
        player.say("Treasure room!")  
player.on_travelled(on_walk)
```

No typing — it fires every time the player walks.

이벤트 = 어떤 일이 생기면 자동 실행

CONCEPT

Different events, different triggers



on_travelled

fires when the
player walks



on_block_placed

fires when the
player builds

행동마다 다른 이벤트가 있어요

CONCEPT

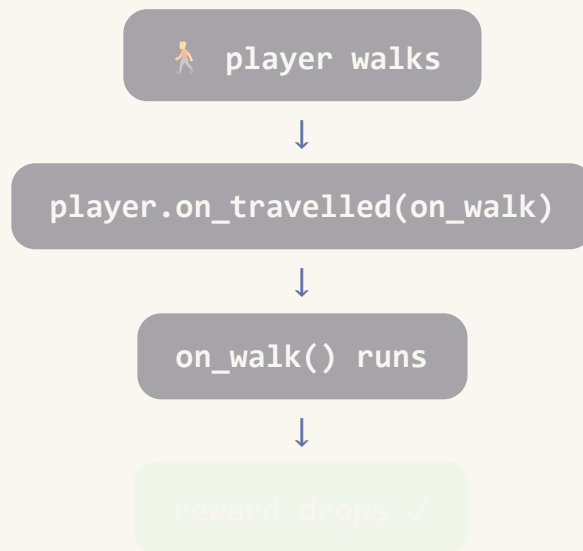
Event → call a function

```
def drop_reward():  
    agent.place(DIAMOND_BLOCK, DOWN)  
  
def on_walk():  
    if agent.inspect(DOWN) == REDSTONE_BLOCK:  
        drop_reward()  
player.on_travelled(on_walk)
```

이벤트 안에서 함수를 호출해요

WATCH IT!

Walk → fires the event



걷는 순간 이벤트가 실행돼요

PREDICT 🤔

How many messages?

Predict first — then click reveal

```
# player walks: stone, stone, gold, stone
def on_walk():
    if agent.inspect(DOWN) == GOLD_BLOCK:
        player.say("Treasure room!")
player.on_travelled(on_walk)
```

Reveal output 🔍 reveal

VOCAB

New words today

event code that runs by itself on an action

trigger the action that makes the event fire

on_travelled event for when the player walks

flag a variable that remembers if done

이벤트

방아쇠

걸을 때

기록 변수



What's wrong with this code?

Goal: the reward fires **once** on the start tile — but it floods diamonds.

buggy

```
def on_walk():  
    if agent.inspect(DOWN) == REDSTONE_BLOCK:  
        agent.place(DIAMOND_BLOCK, DOWN)  
player.on_travelled(on_walk)
```

It fires on every step on redstone.



The bug

Bug: nothing remembers the reward was given, so it repeats. Use a `flag` variable.

fixed

```
reward_given = False
def on_walk():
    global reward_given
    if agent.inspect(DOWN) == REDSTONE_BLOCK and not reward_given:
        agent.place(DIAMOND_BLOCK, DOWN)
        reward_given = True
player.on_travelled(on_walk)
```

YOUR TURN!

Build a mini game zone

- ✓ One trap, one reward
- ✓ Each fires from a player event
- ✓ Check the block before reacting
- ✓ Make the reward fire only once
- ✓ Send a video of both triggering



QUIZ

QUESTION 1 OF 3

How is an **event** different from a chat command?

It runs only once ever.

There's no difference.

It can only place blocks.

It runs by itself on an action — you don't type anything.



QUIZ

QUESTION 2 OF 3

A reward keeps firing every step. Best fix?

Make the player walk slower.

Use a flag variable to remember it was given.

Remove the event entirely.

Place more diamonds.



QUIZ

QUESTION 3 OF 3

A "Nice building!" message should fire when the player **builds**. Which event?

`on_travelled`

`agent.move`

`on_chat`

`on_block_placed`

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Your world reacts!

Next: build your function library.

다음 주: 나만의 함수 라이브러리